# Zeroth-Order Blind Interference Suppression for Multi-RIS-Aided Wireless Systems

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***Abstract*—In this paper, we study measurement-driven signal-to-interference-plus-noise ratio (SINR) maximization for a multi-reconfigurable intelligent surface (RIS)-aided single-input single-output (SISO) system with unknown strong interference sources. Specifically, the objective is to optimize the reflection coefficients such that the SINR is maximized at the receiver. As the interference channels are unknown, such an optimization problem is a black-box optimization problem with an objective function whose closed-form analytical expression is unknown. To address the high-dimensional black-box optimization problem with discrete variable constraints, we introduce a group-based phase parameterization that significantly reduces the search dimension. Building on this model, we develop a group-based zeroth-order adaptive moment (ZO-AdaMM) algorithm. Simulation results show that the proposed grouping strategy markedly accelerates the convergence speed and achieves a superior interference suppression performance under limited measurement budgets, especially in the small-budget regime.**

***Index Terms*—Reconfigurable intelligent surface (RIS), zeroth-order optimization, CSI-free, discrete phase shift, interference suppression.**

## I. Introduction

Reconfigurable intelligent surface (RIS) has been a paradigm shift for wireless systems [1], [2] due to their capability to reshape wireless propagation through programmable phase shifts, thereby enabling capacity/coverage enhancement [3], interference mitigation [4], and energy-efficient transmission [5]. Deploying multiple RISs is essential to improve system throughput and robustness by providing flexible and controllable propagation paths [6]. However, acquiring channel state information (CSI) is inherently challenging in RIS-assisted systems [7], and this difficulty is further exacerbated by the increasing number of RISs [8]. This is due to the multiple reflection links introduced by two or more RISs, especially in the presence of non-cooperative interferences.

Recently, blind beamforming without explicit CSI acquisition has emerged as a promising solution [9]–[12]. In this approach, the reflection coefficients of the RIS are configured by collecting a large number of received signal power measurements and optimizing each coefficient based on maximizing the conditional expectation. Specifically, the authors in [9] developed multi-RIS sequential conditional sample mean (SCSM) method to estimate the conditional mean for each discrete phase shift state and update phase sequentially. SCSM can achieve a quartic $\mathcal{O}(M^4)$ signal-to-noise ratio (SNR) gain in a double-RIS system under favorable rank-one/line-of-sight (LoS) conditions and its benefits have been validated in prototypes, where $M$ is the number of reflected elements. Nevertheless, CSM-based methods rely on a large number of samples to approximate the conditional expectation, which can lead to a prohibitively high sample complexity.

To efficiently leverage the inherent structure of the received signal power, the work in [4], [11] address the problem of blind interference suppression by proposing to implicitly infer the null space of the interference covariance matrix based solely on received signal power measurements. Although achieving superior performance, the proposed method requires control of the reflection amplitudes as well as the phase shifts, both of which are assumed to be continuous variables, and performance degradation may occur otherwise. To alleviate the hardware burden of RIS with phase-shift-only control, we introduced a derivative-free approach in [12] that treats blind interference suppression as a black-box problem [13] and relies solely on received signal power measurements to construct an objective function that maximizes the signal-to-interference-plus-noise ratio (SINR).

In this paper, we consider a more challenging blind interference suppression scenario involving multi-RIS with discrete phase shifts only. To further reduce hardware complexity and accelerate convergence, we propose a group-based zeroth-order adaptive moment method (ZO-AdaMM). By grouping adjacent elements to reduce the effective dimension, the proposed method achieves faster, more sample-efficient convergence and higher SINR in the small-budget regime, as revealed by simulation results.

## II. System Model

As shown in Fig.1, this paper considers a multi-RIS-assisted SISO system in the presence of $K$ unknown strong interference signals. By leveraging the channel reconfiguration capabilities of RISs, we aim to suppress the unknown interference signals while simultaneously enhancing legitimate communication performance.

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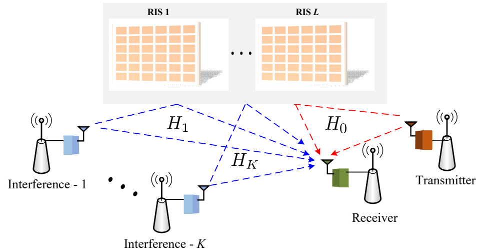


Fig. 1. The RISs-assisted wireless communication system.

The $l$th RIS employs a uniform planar array (UPA) structure with $M_l = M_{l,x} \times M_{l,y}$ reflecting elements, and its reflection phase shift matrix is given by

$$\boldsymbol{\Theta}_l \triangleq \operatorname{diag}(e^{j\boldsymbol{\theta}_l^H}) \tag{1}$$

where $\boldsymbol{\theta}_l = [\theta_{l,1} \ \ \ldots \ \ \theta_{l,M_l}]^H$ denotes the reflection phase shift vector, and $\theta_{l,m}$ denotes the phase shift of the $m$th element at the $l$th RIS. In practice, the phase shift often is selected from a finite set of discrete values due to hardware limitations. With $B$ quantization bits, each phase value can be represented by one of $2^B$ discrete states, i.e., there are $|\mathcal{F}| = 2^B$ admissible phase levels with quantization step defined as $\Delta \triangleq \frac{2\pi}{2^B}$. With B quantization bits, each phase shift $\theta_{l,m}$ can be selected from a set of discrete states; i.e., there are

$$\mathcal{F} \triangleq \{0, \Delta, 2\Delta, \ldots, (2^B - 1)\Delta\}. \tag{2}$$

By stacking all $L$ reflecting phase shift vectors, the aggregated phase-shift vector of all RIS elements is defined as

$$\boldsymbol{\theta} \triangleq [\boldsymbol{\theta}_1^H \ \ \ldots \ \ \boldsymbol{\theta}_L^H]^H \tag{3}$$

In general, the composite channel from the transmitter to the receiver comprises the direct link, the reflected links via a single RIS, and potential multi-hop reflected links via multi-RIS, as illustrated in Fig. 1. To sidestep the intricate channel modeling associated with RISs, we define $H_k(\boldsymbol{\theta}), k = 0, 1, \ldots, K$ as the effective end-to-end channel from the legitimate transmitter ($k = 0$) and the $k$th interference source ($k = 1, 2, \ldots, K$) to the receiver.

Clearly, the received signal at the receiver is the superposition of the legitimate signal and interference signals, i.e.,

$$y(t) = H_0(\boldsymbol{\theta})s(t) + \sum\nolimits_{k=1}^{K} H_k(\boldsymbol{\theta})j_k(t) + \epsilon(t), \tag{4}$$

in which $s(t) \sim \mathcal{CN}(0, \sigma_0^2)$ is the transmitted signal from the legitimate transmitter, $j_k(t) \sim \mathcal{CN}(0, \sigma_k^2)$ denotes the interference signal from the $k$th interferer, and $\epsilon(t) \sim \mathcal{CN}(0, \sigma_w^2)$ is additive white Gaussian noise.

In this paper, we consider block fading channels where the channel remains constant over a given coherence time block. In addition, the desired signal and the interference signals are assumed to be mutually uncorrelated.

Since the interference signals and noise are unknown at the receiver, we obtain quadratic measurements by averaging over sufficiently many samples to capture their second-order statistics. Specifically, at the receiver, only the average received signal power can be observed, which is given by

$$\begin{aligned} P(\boldsymbol{\theta}) &\triangleq \mathbb{E}[|y(t)|^2] \qquad (5)\\ &= \underbrace{\mathbb{E}[|H_0(\boldsymbol{\theta})s(t)|^2]}_{P_S(\boldsymbol{\theta})} + \underbrace{\mathbb{E}\left[\left|\sum\nolimits_{k=1}^{K} H_k(\boldsymbol{\theta})j_k(t)\right|^2\right]}_{P_I(\boldsymbol{\theta})} + \underbrace{\sigma_w^2}_{P_N} \\ &= \sigma_0^2\mathbb{E}[|H_0(\boldsymbol{\theta})|^2] + \sum\nolimits_{k=1}^{K} \sigma_k^2\mathbb{E}[|H_k(\boldsymbol{\theta})|^2] + \sigma_w^2. \qquad (6) \end{aligned}$$

The received SINR directly using only received power observations, which is defined as

$$\mathrm{SINR}(\boldsymbol{\theta}) \triangleq \frac{P_S(\boldsymbol{\theta})}{P_I(\boldsymbol{\theta}) + \sigma_w^2}. \tag{7}$$

For legitimate transmission, the desired signal power $P_S(\boldsymbol{\theta})$ can be calculated using effective channel function $H_0(\cdot)$, which is acquired via pilot-aided estimation methods [10], [11]. Since the total received power $P(\boldsymbol{\theta})$ can be directly measured and $P_S(\boldsymbol{\theta})$ is known, the SINR in (7) can be equivalently evaluated as

$$\mathrm{SINR}(\boldsymbol{\theta}) = \frac{P_S(\boldsymbol{\theta})}{P(\boldsymbol{\theta}) - P_S(\boldsymbol{\theta})}. \tag{8}$$

## III. Problem Formulation

In this paper, the objective is to maximize the received SINR by optimizing the RIS phase shift vector without any knowledge of the CSI relevant to interference sources. The optimization problem is formulated as

$$\begin{aligned} (\mathrm{P1}) \quad \max_{\boldsymbol{\theta}} \quad & \mathrm{SINR}(\boldsymbol{\theta}) && (9)\\ \text{s.t.} \quad & \theta_{l,m} \in \mathcal{F}, \quad \forall l, m, && (10) \end{aligned}$$

where $\mathrm{SINR}(\boldsymbol{\theta})$ is defined in (7).

Problem (P1) is non-convex and combinatorial due to the fractional objective and the discrete constraints on the RIS reflection coefficients. When the CSI of both the desired and interference links is available, such a problem can be solved by first obtaining a continuous solution via manifold optimization [14] methods and then projecting it onto the discrete set. However, the interference CSI is typically unavailable as the interference sources are inherently non-cooperative. Consequently, the resulting SINR values we can only obtain via measurements as defined in (8). This renders problem (P1) a black-box optimization problem [13], for which traditional CSI-dependent methods are inapplicable.

## IV. Proposed Method

In this section, we present a group-based zeroth-order (ZO) passive beamforming scheme. To reduce the optimization dimension, the RIS elements are partitioned into sub-groups of adjacent reflecting elements, where elements within each group share a common phase shift.

We first develop the algorithm under a continuous phase-shift assumption and subsequently extend it to discrete settings. By leveraging a ZO gradient approximation to determine the search direction, the proposed algorithm iteratively updates the group phase shifts based on SINR measurements. For the discrete case, the updated variables are projected onto a feasible discrete set to comply with practical hardware constraints.

### A. Grouped Reflection Phase Shifts Modeling

As shown in Fig. 2, each RIS is partitioned into $N_{x,l} \times N_{y,l}$ non-overlapping rectangular groups, each with a size of $B_x \times B_y$, where

$$N_{x,l} = \left\lceil \frac{M_{l,x}}{B_x} \right\rceil, N_{y,l} = \left\lceil \frac{M_{l,y}}{B_y} \right\rceil. \tag{11}$$

Let $\boldsymbol{\theta}^{blk} = [\theta_1^{blk} \ \ldots \ \theta_d^{blk}]^T \in \mathbb{C}^{D_l}, D_l = N_{x,l} N_{y,l}$ denote the group phase shift vector comprising common phase shifts among adjacent reflecting elements. The overall phase shift vector $\boldsymbol{\theta}$ can then be recast as

$$\boldsymbol{\theta} = \mathcal{M}(\boldsymbol{\theta}^{\text{blk}}), \tag{12}$$

where $\mathcal{M}(\cdot)$ is a deterministic one-to-many mapping.

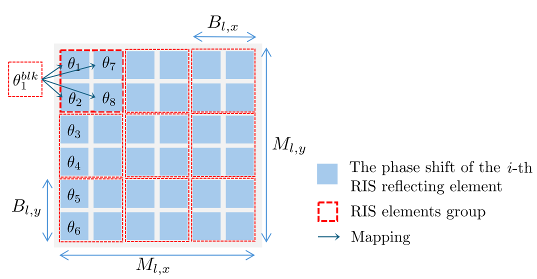


Fig. 2. Grouped reflection phase shifts model.

### B. Group-Based ZO Optimization

To begin with, we propose a **ZO-AdaMM-based algorithm** to solve the resulting continuous black-box problem. By employing the group-based modeling, the effective optimization dimension is reduced to $D = \sum_{l=1}^{L} D_l$.

At the $n$-th iteration, a random direction $\boldsymbol{u}[n] \in \mathbb{R}^D$ is generated, and a central-difference ZO gradient estimator is constructed as

$$\widehat{\nabla} f(\boldsymbol{\theta}^{\text{blk}}[n]) = \frac{D}{2\mu}\Big[f(\boldsymbol{\theta}_+^{\text{blk}}[n]) - f(\boldsymbol{\theta}_-^{\text{blk}}[n])\Big]\boldsymbol{u}[n], \tag{13}$$

where $\mu > 0$ is a smoothing parameter, $\boldsymbol{\theta}_\pm^{\text{blk}}[n] \triangleq \boldsymbol{\theta}^{\text{blk}}[n] \pm \mu \boldsymbol{u}[n]$, and $f(\boldsymbol{\theta}^{\text{blk}}) \triangleq \text{SINR}(\mathcal{M}(\boldsymbol{\theta}^{\text{blk}}))$ denotes the measured SINR value when RISs are configured with $\boldsymbol{\theta} = \mathcal{M}(\boldsymbol{\theta}^{blk})$. The ZO-AdaMM update at the $n$-th iteration is then carried out as

$$\widehat{\nabla} g[n] \triangleq \widehat{\nabla} f(\boldsymbol{\theta}^{\text{blk}}[n]), \tag{14}$$

$$\boldsymbol{m}[n] = \beta_1 \boldsymbol{m}[n-1] + (1-\beta_1)\widehat{\nabla} g[n], \tag{15}$$

$$\boldsymbol{v}[n] = \beta_2 \boldsymbol{v}[n-1] + (1-\beta_2)\widehat{\nabla} g[n] \odot \widehat{\nabla} g[n], \tag{16}$$

$$\hat{\boldsymbol{v}}[n] = \max(\hat{\boldsymbol{v}}[n-1], \boldsymbol{v}[n]), \tag{17}$$

and the group phase shifts vector is updated according to

$$\boldsymbol{\theta}[n+1]^{\text{blk}} = \boldsymbol{\theta}^{\text{blk}}[n] + \alpha[n]\hat{\boldsymbol{V}}[n]^{-1/2}\boldsymbol{m}[n], \tag{18}$$

where $\hat{\boldsymbol{V}}[n] = \text{diag}(\hat{\boldsymbol{v}}[n])$ and $\alpha[n]$ denotes the step size. The proposed grouped ZO method is guaranteed to converge to a critical point [15].

### C. Extension to Discrete Phase Shifts

To enforce the discrete constraint, we define an element-wise quantization (projection) operator $\mathcal{Q}(\cdot)$ as

$$\mathcal{Q}(x) \triangleq \arg\min_{\phi \in \mathcal{F}} |x - \phi|, x \in [0, 2\pi). \tag{19}$$

At the $n$-th iteration, the two probing points used in the central-difference estimator in (13) are quantized as

$$\tilde{\boldsymbol{\theta}}_\pm^{\text{blk}}[n] = \mathcal{Q}\Big(\boldsymbol{\theta}^{\text{blk}}[n] \pm \mu \boldsymbol{u}[n]\Big). \tag{20}$$

The ZO gradient estimator is computed by replacing the two function evaluations in (13) with $f(\tilde{\boldsymbol{\theta}}_+^{\text{blk}}[n])$ and $f(\tilde{\boldsymbol{\theta}}_-^{\text{blk}}[n])$. After each ZO-AdaMM update, the resulting phase vector is projected back onto the feasible set:

$$\tilde{\boldsymbol{\theta}}^{\text{blk}}[n+1] \leftarrow \mathcal{Q}(\boldsymbol{\theta}^{\text{blk}}[n+1]). \tag{21}$$

Notably, quantization may compromise the convergence guarantees of the ZO-AdaMM algorithm. Nevertheless, the simple projection approach still exhibits satisfactory performance, as demonstrated in the following numerical simulation.

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**Algorithm 1:** Group-Based ZO-AdaMM

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**Input:** Quantized group phase vector $\tilde{\boldsymbol{\theta}}^{\text{blk}}[0]$, step size $\alpha$, smoothing parameter $\mu$, momentum parameters $\beta_1, \beta_2$, discrete set $\mathcal{F}$.
**Output:** Optimized discrete phase shifts vector $\tilde{\boldsymbol{\theta}}$.

Initialize $\boldsymbol{m}[0] = \mathbf{0}$, $\boldsymbol{v}[0] = \hat{\boldsymbol{v}}[0] = \mathbf{0}$;
**for** $n = 1, 2, \ldots, N-1$ **do**
  $\tilde{\boldsymbol{\theta}}_\pm^{\text{blk}}[n] = \mathcal{Q}\Big(\tilde{\boldsymbol{\theta}}^{\text{blk}}[n] \pm \mu \boldsymbol{u}[n]\Big)$;
  $\widehat{\nabla} g[n] = \dfrac{D}{2\mu}\Big(f(\tilde{\boldsymbol{\theta}}_+^{\text{blk}}[n]) - f(\tilde{\boldsymbol{\theta}}_-^{\text{blk}}[n])\Big)\boldsymbol{u}[n]$;
  $\boldsymbol{m}[n+1] = \beta_1 \boldsymbol{m}[n] + (1-\beta_1)\widehat{\nabla} g[n]$;
  $\boldsymbol{v}[n+1] = \beta_2 \boldsymbol{v}[n] + (1-\beta_2)\widehat{\nabla} g[n] \odot \widehat{\nabla} g[n]$;
  $\hat{\boldsymbol{v}}[n+1] = \max\big(\hat{\boldsymbol{v}}[n], \boldsymbol{v}[n+1]\big)$;
  $\boldsymbol{\theta}^{\text{blk}}[n+1] = \tilde{\boldsymbol{\theta}}^{\text{blk}}[n] + \alpha \hat{\boldsymbol{V}}[n+1]^{-1/2}\boldsymbol{m}[n+1]$;
  $\tilde{\boldsymbol{\theta}}^{\text{blk}}[n+1] = \mathcal{Q}\Big(\boldsymbol{\theta}^{\text{blk}}[n+1]\Big)$;
**end**
**return** $\tilde{\boldsymbol{\theta}}^{\text{blk}}[N]$, $\tilde{\boldsymbol{\theta}}[N] = \mathcal{M}(\tilde{\boldsymbol{\theta}}^{\text{blk}}[N])$;

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## V. Simulation Results

In this section, we evaluate the performance of the proposed group-based ZO-AdaMM method for a double-RIS-assisted communication system [16]. Each RIS comprises

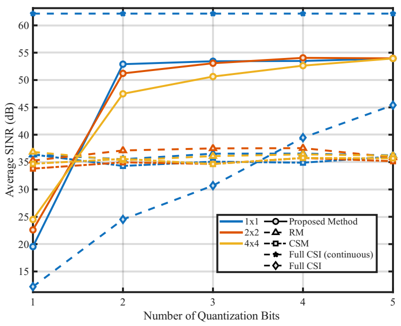


Fig. 3. SINR vs. the number of quantization bits with $T = 2000$.

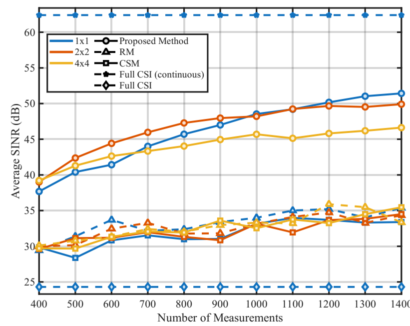


Fig. 4. SINR vs. the number of measurements $T$ for different group sizes.

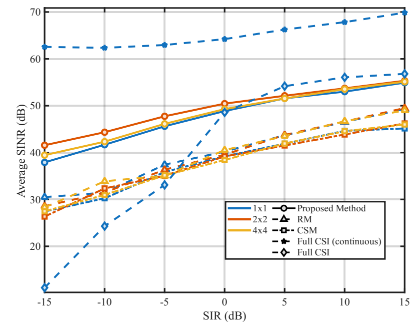


Fig. 5. SINR vs. SIR with a fixed $T = 600$.

$16 \times 16$ reflecting elements ($M_1 = M_2 = 256$), with each element employing a low-resolution 2-bit phase shifter.

The transmitter, receiver, RIS 1, and RIS 2 are located at $(1, 0, 2)$, $(1, 50, 0)$, $(0, -5, 1)$, and $(0, 55, 1)$ meters, respectively. In this work, we focus on the case of a single interference source, i.e., $K = 1$, which is located at $(55, 0, 0)$ meters. Large-scale fading is modeled using distance-dependent path-loss functions with different parameters. Specifically, the path loss for the direct link is $32.6 + 36.7 \log_{10} d$ , while the corresponding path loss for the reflected link is $30 + 22 \log_{10} d$, where $d$ is the distance in meters. The receiver noise power is set to as $\sigma_w^2 \approx -107$ dBm. The signal-to-interference ratio (SIR) is defined as $\text{SIR} \triangleq \sigma_0^2 / \sum_{k=1}^{K} \sigma_k^2$, which is fixed at $-10$ dB and the transmit power of the legitimate transmitter is fixed at 30 dBm. All results are averaged over 500 random channel realizations.

To demonstrate the effectiveness of the proposed method, several state-of-the-art benchmark schemes are considered for comparison:

- **RM(RandomMax):** Specifically, $T$ candidate group phase shift vectors are randomly sampled from the discrete set $\mathcal{F}$ and the one that yields the maximum observed SINR is selected.
- **CSM(Conditional Sample Mean [9]):** After randomly generating $T$ group phase shift vectors and obtaining the corresponding SINR values, each phase shift is selected by maximizing the conditional expectation.
- **Full CSI (continuous):** Assuming perfect CSI for all channels, problem (P1) is solved using manifold optimization [14]. This approach serves as a performance upper bound for the proposed method.
- **Full CSI:** Based on the Full CSI (continuous), the optimized reflection coefficients are further projected onto the discrete set.

Fig. 3 illustrates the average SINR as a function of the number of quantization bits $B$ with a sufficiently large number of measurements $T = 2000$. As expected, the full CSI (continuous) method provides an upper bound on the performance of all competing schemes. In contrast, the full-CSI approach experiences significant performance degradation when projecting onto the discrete phase set directly. The proposed method is inferior to the RM and CSM methods under 1-bit quantization but outperforms them by a significant margin as the number of quantization bits increases across different group sizes. The primary reason is that the gradient approximation in ZO-AdaMM is severely degraded under 1-bit quantization, whereas 2-bit resolution suffices to enable effective gradient estimation and realize substantial beamforming gains. Furthermore, the performance gap between different group sizes diminishes as the phase resolution increases to 5 bits, indicating that 2-bit phase shifters offer an optimal trade-off between hardware complexity and system performance for the considered scenario.

Fig. 4 illustrates the average SINR as a function of the number of the measurements $T$ for different group sizes. Both the proposed method and the CSM method achieve higher SINR performance as the number of measurements $T$ increases. Moreover, the proposed method significantly outperforms the RM and CSM methods by a remarkable margin. In addition, it can be observed that, under a limited number of measurements (e.g., $T \leq 700$), a larger group size yields a higher SINR compared to a smaller group size, demonstrating a clear advantage in practical scenarios.

Fig. 5 depicts the average SINR versus SIR with a fixed number of measurements of $T = 600$. The proposed method consistently outperforms RM and CSM across the entire SIR range for all tested group sizes. Moreover, the benefits of the proposed grouping method remain evident under moderate and strong interference conditions.

## VI. Conclusion

In this paper, we developed a group-based ZO-AdaMM in which adjacent reflecting elements share a common phase shift through discrete phase projection. Simulation results demonstrated that the proposed approach achieves superior performance in terms of both SINR improvement and sample efficiency, making it well-suited for resource-constrained multi-RIS deployments.

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